Japan Cherry Blossom Blooming Data Report

Introduction

The timing of cherry blossom flowering and full bloom is linked to environmental conditions such as temperature. As such, we can associate flowering times with recent climate variability and warming patterns. While previous observations suggest that bloom timings may be occurring earlier in the year due to climate change, the extent of this change across different geographic contexts may require further inquiry due to factors such as regional location and climate classification.

This report aims to analyze cherry blossom full bloom data across multiple weather stations in Japan using Python data processing tools. It aims to explore the following objectives:

1. Assess long-term trends in full bloom timing.
2. Determine the typical timing and variability of full bloom dates.
3. Identify years with extremely early or late full bloom timings.
4. Examine differences in full bloom timing across different climate classifications and provinces.

Methodology

I. Data Sources

This project uses multiple datasets. The primary dataset consists of historical cherry blossom observations obtained from a Kaggle dataset, “Japan Cherry Blossom Observation”. It includes periodic records of events such as “flowering” and “full bloom” at various observation stations. Station metadata were obtained from the “World Meteorological Organization (WMO) Station List”, while additional geolocation information was retrieved using the Nominatim API. Historical climate-related data were obtained from Open-Meteo API and the Meteostat Python package.

II. Data Processing and Integration

Data processing and analysis were implemented using Python with the following libraries: Pandas, numpy. The initial cleaned dataset was generated by merging the raw cherry blossom observation data (blossom.csv) with station metadata (stations.csv) using a common station identifier. From here, the dataset was further enriched by merging geolocation data such as province and climate classification.

III. Climate Classification

Climate conditions at each station were categorized using the Köppen climate classification system. This system groups regions based on temperature and precipitation patterns (Peel et al., 2007). The classifications observed in this study may be generalized as follows:

- Af: High temperatures and rainfall
- Cfa: Hot summers, mild winters, and humidity
- Dfa: Hot summers and cold winters
- Dfb: Mild summers and very cold winters

IV. Data Analysis

All analyses were implemented using Python scripts and exported to the “/outputs” directory.

Temporal trends were assessed by calculating the annual average day of year for full bloom and using a linear regression model to estimate the rate of change over time. The typical bloom timing was evaluated using mean, median, and the distribution of full bloom dates. Extreme events were identified by determining the minimum and maximum day of year values, meanwhile variability in bloom timing was assessed using standard deviation and range to evaluate consistency over time.

To evaluate environmental influences, bloom timing was compared across Köppen climate classifications. Trends were also computed for each classification to observe differences. Geographic variations were also analyzed at the provincial level by comparing bloom timing and trends across different provinces.

V. Visualization

Visualization was implemented using the matplotlib library. Time-series graphs were used to show trends over time, histograms to show distributions of bloom timings, and boxplots to compare distributions across climate classes and regions.

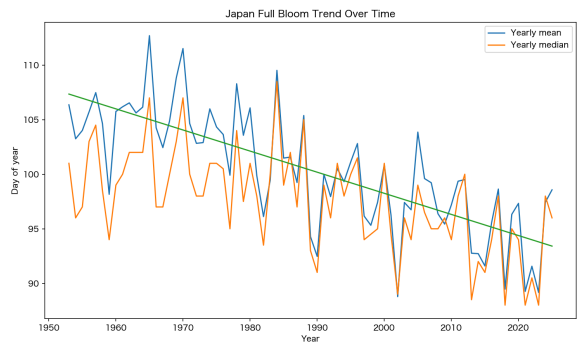
VI. Code Availability

All scripts and processed data used in this project are available at: https://github.com/jellyM0on/cttc-cherry-blossom

Results & Discussion

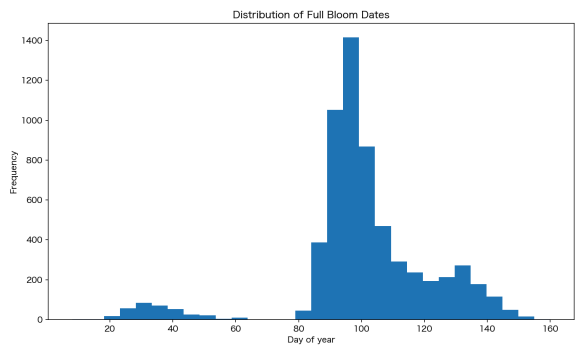
I. Long-term Trends

Results show a long-term shift toward earlier cherry blossom full bloom timing in Japan. The national trend exhibits a negative slope of approx. -0.19 days per year (mean). This indicates events are occurring progressively earlier. This supports the hypothesis that blooming is happening earlier, likely due to environmental changes like rising temperatures as a result of global warming.



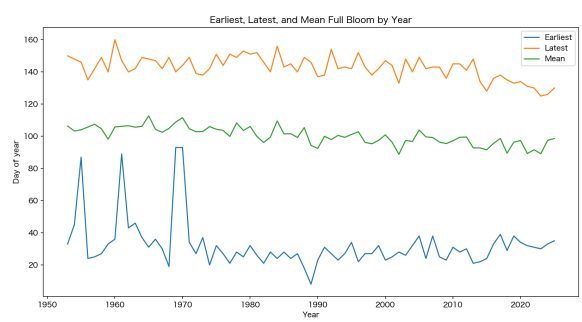
II. Typical Timing and Distribution

Full blooming typically occurs around day 100 of the year. Most bloom dates fall within the range day 93 to 110, around the month of April. However, certain regions have very early dates within the range day 20 to 60. This is particularly the case for regions such as Okinawa (Af), highlighting the importance of geographical factors.



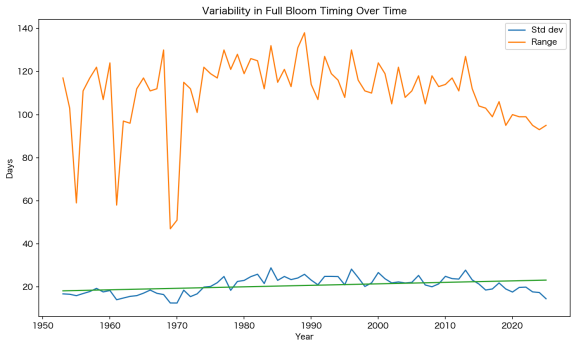
III. Extremities

The earliest recorded full bloom occurred in 1989 in Okinawa (Af) while the latest recorded full bloom occurred in 1960 in Hokkaido (Dfb). This lines up with geographical patterns. Warmer tropical climates tend to produce earlier blooms. Colder regions produce later blooms.



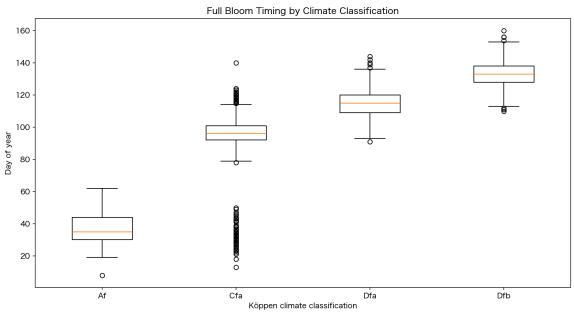
IV. Variability and Consistency

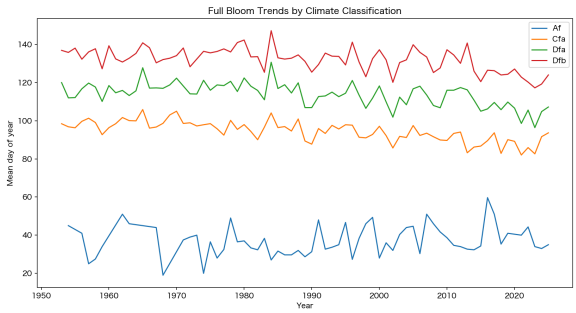
Although it can be observed that there is a trend leaning towards a slight increase in bloom timing, data from recent years indicate a potential decline in both the standard deviation and range. This suggests that, at one point, a period of stabilization has occurred. This may be attributed to increasingly uniform environmental conditions across regions. For instance, due to current climate trends, regions may now be getting consistently warmer.



V. Difference Across Climate Classifications

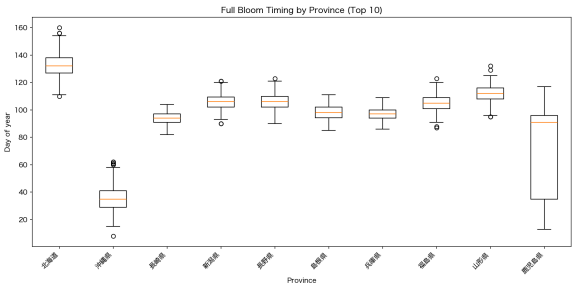
The distribution of full bloom timing across the climate classifications reveal that the Cfa climate shows a significantly higher number of outliers. A substantial number of earlier and later bloom observations are present. This likely reflects the fact that this climate classification covers much of central and southern Japan, including coastal and inland regions. This climate class intrinsically has more mixed characteristics than tropical and continental climates, hence we can observe this kind of variability. Still, overall the common pattern that tropical regions (Af) experience earlier blooms and colder (Dfa/Dfb) regions experience later blooms still holds.

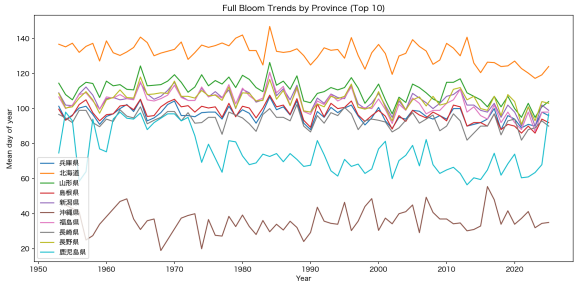




VI. Provincial Differences

Exploring the provincial differences shows us a clear geographic pattern. Far southern regions such as Okinawa (沖縄県) tend to have earlier blooms. Northern regions like Hokkaido (北海道) tend to have later blooms. Other regions that experience comparatively less extremities like Nagasaki (長崎県) and Shimane (島根県) tend to have average timing blooms. Trends seem consistent over time for most





regions, except for Kagoshima (鹿児島県) which shows a sudden upward spike for recent years. This can be associated with how Kagoshima is frequently affected by weather anomalies, primarily due to its position as a "typhoon gateway" where warm ocean currents fuel intense rainfall. This is also further intensified by Mt. Sakurajima's volcanic activity, as it generates unique ash-storms that disrupt the atmosphere and create localized environmental conditions found nowhere else in the country (Rahadianto et al., 2022).

VII. Synthesis

- Bloom dates are occurring earlier over time on average. This suggests temperatures are consistently getting warmer.
- Climates and geography strongly influence cherry blossom timing. Warmer regions tend to have earlier blooms.
- Despite average trends, there is still substantial variability among different Japan regions, but recent patterns seem to suggest a potential declining trend in the future.

References

Peel, M. C., Finlayson, B. L., & McMahon, T. A. (2007). Updated world map of the Köppen-Geiger climate classification. Hydrology and Earth System Sciences, 11(5), 1633–1644. https://hess.copernicus.org/articles/11/1633/2007/

Rahadianto, H., Tatano, H., Iguchi, M., Tanaka, H. L., Takemi, T., & Roy, S. (2022). Long-term ash dispersal dataset of the Sakurajima Taisho eruption for ashfall disaster countermeasure. *Earth System Science Data*, *14*(11), 5309–5332. https://doi.org/10.17603/ds2-vw5f-t920